

Van-Loc Nguyen

AI Engineering, Autonomous, Ho Chi Minh City, Vietnam

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RESEARCH INTERESTS

Computer Vision, Deep Learning, Natural Language Processing, Multimedia Verification

EDUCATION

University of Science, VNUHCM, Ho Chi Minh City, Vietnam

Dec 2024 — Present

Master of Science in Computer Science, majoring in Artificial Intelligence

University of Science, VNUHCM, Ho Chi Minh City, Vietnam

Oct 2024 — Oct 2024

Bachelor of Science in Computer Science (Honors Program)

GPA: 3.88/4.00

Thesis Title: Fake News Analysis and Detection

EXPERIENCE

Autonomous

Ho Chi Minh City, Vietnam

AI Engineer

Mar 2025 — Present

- Be responsible for building fantastic AI agents for real-world needs.
- Improve existing MCP servers to enhance the user experience.
- Optimize AI agents to run on a local environment with a small model (for example, 4B models).

SELab, University of Science, VNUHCM

Ho Chi Minh City, Vietnam

Research Assistant

Jan 2025 — Present

- Participate in research related to multimedia verification.
- Currently proposing solutions for some challenges at ACM Multimedia 2025.

Saritas Vietnam

Ho Chi Minh City, Vietnam

Backend Developer

Mar 2023 — Feb 2025

- Developed, optimized, and maintained the backend side for several projects, with a significant number of users in the US market, in Django and Django REST Framework, combined with PostgreSQL and Celery.
- Familiar with Version Control System, especially Git, be aware of Git flow and GitHub flow.
- Experienced with DevOps tools such as Kubernetes or Teleport.

PUBLICATIONS

- Gia-Huy Vuong, **Van-Loc Nguyen**, Van-Son Ho, Tien-Thanh Nguyen Dang, Ngoc-Do Tran, Van-Tu Ninh, and Minh-Triet Tran (2025). NewsInsight2.0: An Enhanced Version Integrating Large Language Model-Based Query Optimization with Advanced Temporal Mechanisms. In: *Information and Communication Technology. SOICT 2024. Communications in Computer and Information Science, vol 2353*. Springer, Singapore, DOI: 10.1007/978-981-96-4291-5_28.
- Thanh-Son Nguyen*, Viet-Tham Huynh*, **Van-Loc Nguyen***, and Minh-Triet Tran (2024). An Approach to Complex Visual Data Interpretation with Vision-Language Models. In *Proceedings of the Asian Conference on Computer Vision (ACCV) Workshops* (pp. 334-350), DOI: 10.1007/978-981-96-2641-0_23.
- Minh-Triet Tran, Bao-Tin Nguyen, **Van-Loc Nguyen**, Thanh-Son Nguyen, Mai-Khiem Tran and Trong-Le Do (2024). AI-Enhanced Photo Authenticity: A User-Focused Approach to Detecting and Analyzing Manipulated Images. *2024 International Conference on Multimedia Analysis and Pattern Recognition (MAPR)*, Da Nang, Vietnam, 2024, pp. 1-6, DOI: 10.1109/MAPR63514.2024.10661051.
- Bao-Tin Nguyen*, **Van-Loc Nguyen***, Thanh-Son Nguyen, Duc-Tien Dang-Nguyen, Trong-Le Do, and Minh-Triet Tran. 2024. A Hybrid Approach for Cheapfake Detection Using Reputation Checking and End-To-End Network. In *Proceedings of the 1st Workshop on Security-Centric Strategies for Combating Information Disorder (SCID '24)*. Association for Computing Machinery, New York, NY, USA, Article 3, 1–12. DOI: 10.1145/3660512.3665521.
- **Van-Loc Nguyen***, Bao-Tin Nguyen*, Thanh-Son Nguyen, Duc-Tien Dang-Nguyen, and Minh-Triet Tran. 2024. A Unified Network for Detecting Out-Of-Context Information Using Generative Synthetic Data. In *Proceedings of the 2024 International Conference on Multimedia Retrieval (ICMR '24)*. Association for Computing Machinery, New York, NY, USA, 1300–1305. DOI: 10.1145/3652583.3657599.

PROJECTS

Fake News Analysis and Detection

Ho Chi Minh City, Vietnam

Undergraduate thesis at the University of Science, VNUHCM.

Main contributions:

- Propose the usage of an end-to-end network of Visual Entailment task to solve both 2 tasks of cheapfakes detection.
- Cheapfakes data augmentation method with LLMs.
- Reputation checking the image over the Internet to utilize social information.
- An open system to assist users in fake news analysis and detection with the help of an external forensic framework.

The results of this thesis are published at ICMR 2024, SCID 2024, and MAPR 2024.

Demo repo for thesis [here](#).

AWARDS

Third Prize, The 26th Student Scientific Research Award - Euréka 2024

Ho Chi Minh City, Vietnam

Student Scientific Research Award – Euréka is a prestigious award for students interested in scientific research across the country, contributing to the promotion of the scientific research movement of university students. The Ho Chi Minh Youth Union of Ho Chi Minh City hosts the award in collaboration with the National University of Ho Chi Minh City. Dec 2024

Third Prize, The 2024 “AI City Challenge” (AIC) of Ho Chi Minh City

Ho Chi Minh City, Vietnam

The 2024 AI City Challenge is hosted by the Department of Information and Communications of Ho Chi Minh City, aiming to provide information retrieval solutions on Vietnamese news data. Dec 2024

Certificate of Satisfactory Progress from President of Vietnam National University Ho Chi Minh City

Ho Chi Minh City, Vietnam

This certificate is awarded by the President of Vietnam National University Ho Chi Minh City to students who have graduated with a high distinction degree. Dec 2024

ENGLISH

IELTS (Academic): 7.0 (overall score - 2019).

- Listening: 7.0 — Reading: 9.0
- Writing: 6.5 — Speaking: 5.5

SKILLS

- **Programming:** Python, PyTorch, FastAPI, C++.
- **Soft Skills:** Teamwork, communication.